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## HWRS 640 – Computational Methods for Data Driven Earth Science

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**Final Project - Proposal Pitch**

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**Motivation and scientific question** What Earth science problem are you addressing? Why is it interesting or important? State your driving question concisely (1-3 sentences).

Climate-related floods, droughts, and wildfires are intensifying across South America, threatening ecosystems, infrastructure, and vulnerable populations. Recent continental-scale studies have developed multi-hazard susceptibility maps using Earth observation data and spatial multi-criteria decision analysis. However, these approaches rely on static inputs and linear combinations of variables, limiting their ability to capture non-linear relationships and complex spatial interactions. As a result, they provide relative susceptibility but lack predictive capability. This project asks: *Can a data-driven deep learning model improve the identification of hazard hotspots by learning nonlinear spatial relationships from Earth observation data?*

**Dataset** Describe the data you plan to use: What it is (variable(s), domain, temporal/spatial resolution)? Where it comes from (source, how you will access it)? Any known challenges (missing data, large file sizes, preprocessing needed)?

This project will use a comprehensive set of multi-source Earth observation and geospatial datasets representing climatic, hydrologic, topographic, vegetation, and anthropogenic factors, following the structure of recent continental-scale multi-hazard assessments. The input variables include:

All datasets are publicly available and will be accessed primarily through Google Earth Engine and associated repositories. Following the baseline South America framework, all layers will be reprojected to WGS 84, clipped to the South American continent, and harmonized to a common spatial resolution of approximately 1 km. Time-varying predictors will initially be aggregated over a common analysis period (2015–2020) to ensure consistency across the continental domain.

A key challenge is the heterogeneity of the datasets, including differences in spatial resolution (ranging from tens of meters to several kilometers), temporal frequency (daily to annual), and data formats. To address these challenges, preprocessing will include reprojecting, resampling, temporal aggregation, and masking to ensure consistency across all layers. These operations will be performed using `xarray` for handling multi-dimensional arrays and `dask` for scalable, parallel computation, enabling efficient processing of large geospatial datasets.

Table 1: Summary of datasets used in the analysis

| Category                         | Dataset and Description                                                                                                 |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Precipitation                    | CHIRPS dataset for average annual rainfall and extreme daily rainfall                                                   |
| Evapotranspiration               | MODIS MOD16A2 product                                                                                                   |
| Soil Moisture                    | SMAP SPL3SMP_E dataset                                                                                                  |
| Land Surface Temperature         | MODIS MOD11A2                                                                                                           |
| Vegetation Condition             | MODIS NDVI (MOD13A2) and derived indices such as VCI and LAI                                                            |
| Topography                       | MERIT DEM for elevation, slope, and Topographic Wetness Index (TWI)                                                     |
| Hydrologic Features              | Distance to rivers derived from DEM analysis and water occurrence probability from the JRC Global Surface Water dataset |
| Soil Properties                  | OpenLandMap soil texture classification (USDA system)                                                                   |
| Depth to Bedrock                 | Global regolith thickness dataset                                                                                       |
| Land Use and Impervious Surfaces | ESA WorldCover dataset                                                                                                  |
| Fire Data                        | MODIS MCD64A1 burned area product and fire occurrence data from INPE Terrabrisilis                                      |
| Socioeconomic Data               | Population density from WorldPop, and gridded GDP and Human Development Index (HDI)                                     |

**Proposed method** Describe the computational approach you plan to apply: Which method(s) from the course will you use, and why are they appropriate for this problem? What will the inputs and outputs of your model or analysis be? If you are using deep learning, describe the architecture you have in mind (even if preliminary).

As a baseline for comparison, this study follows the structure of existing multi-hazard susceptibility frameworks based on spatial multi-criteria decision analysis. In these approaches, input variables are classified into ordinal susceptibility classes using quantile-based thresholds (e.g., 0–20%, 20–40%, ..., 80–100%). Variables that are expected to reduce susceptibility are directionally transformed to ensure consistent interpretation across criteria. Hazard-specific weight vectors are then applied to combine the classified variables into composite susceptibility indices.

While this framework provides an interpretable and consistent approach across large spatial domains, it relies on discretization and linear weighting, which may oversimplify nonlinear interactions among environmental drivers. In this study, the deep learning model is designed to replace these steps by learning continuous, nonlinear relationships directly from the data.

Therefore, I propose a U-Net style fully convolutional neural network to model continental-

scale spatial patterns of flood, drought, and fire susceptibility across South America. All predictor datasets will be harmonized to a common grid and stacked as multi-channel raster inputs. The model will use an encoder-decoder architecture with skip connections, which allows it to capture large-scale spatial context while preserving local features such as river proximity, topographic gradients, land cover transitions, and urbanized areas.

The encoder will learn shared spatial representations from climatic, hydrologic, topographic, vegetation, and anthropogenic inputs, while separate output heads will generate pixel-wise probability maps for flood, drought, and fire hotspots. This multi-task design is appropriate because the three hazards share some environmental controls but also respond to distinct physical drivers. The model inputs will therefore be raster tiles over South America, and the outputs will be three spatially explicit susceptibility maps at the same grid resolution.

This architecture is better suited than a standard CNN classifier because the objective is not to assign a single label to an image, but to predict hazard probability for every pixel. Compared with the current linear susceptibility framework, the U-Net can learn nonlinear interactions among variables and represent spatial dependencies directly from the gridded data.

**Expected outcomes and evaluation** What do you expect to find? How will you know if your method worked? Describe at least one concrete evaluation metric or diagnostic plot you plan to produce.

The proposed model is expected to improve the identification of hazard-prone regions by capturing nonlinear spatial interactions that are not represented in traditional susceptibility frameworks. Model performance will be evaluated using the Area Under the Receiver Operating Characteristic Curve (AUC), allowing direct comparison with existing methods. Additional evaluation will include spatial comparison of predicted hazard maps with observed hazard datasets and visual assessment of hotspot patterns.

Overall, this project aims to demonstrate that deep learning can enhance multi-hazard assessment by moving from static, linear susceptibility mapping toward data-driven, spatial prediction.